Michigan Tech

Homoclinic Tangles

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Adaptive RK solvers: comparison problem

Test problem: Rössler system together with its 3×3 sensitivity matrix,

$$\dot{y} = f(y), \qquad \dot{J} = Df(y)J.$$

- Locate crossings of a Poincaré section. Save state & sensitivity at crossings.
- Adaptive explicit RK 5(4) solvers with comparable tolerances.
- $3 + 9 = 12$ scalar dynamical variables. Full trajectory not required.
- Julia `Tsit5`, MATLAB `ode45`, SciPy `RK45`, Mma explicit RK in `NDSolve`.

Important

The following are **educated guesses, not benchmarks**. They are intended to indicate plausible relative costs for this particular small ODE/event problem.

Approximate timings. Speed-aware Julia Normalization

System	Simple	Speed-aware
Julia	1.5–2	**1.0**
Mathematica	1.5–2.5	1.1–1.6
MATLAB	2–3.5	1.5–2.5
Python/SciPy	4–8	2–4

NOT measured timings

Deliberately broad **estimates**. Solver tolerances, event frequency/implementation, dense output, compiler versions, and hardware can easily change things substantially.

What “speed-aware” means

Julia. `StaticArrays`: `SVector{3}` for y and `SMatrix{3,3}` for J. Use `Tsit5`, callbacks for the section, and `push!` into a concretely typed vector with optional `sizehint!`.

Mathematica. Event handling for section crossings and minimal interpolation. Data accumulated event records using $O(1)$ construction with post-processing `Flatten` or `Sow`/`Reap` construction.

MATLAB. `ode45` event detection avoiding unnecessary output. Preallocated section data storage. Solver state needs to be a vector sensitivity variables need to be reshaped.

Python/SciPy. `solve_ivp` with `RK45` and an event function. Append section records to a Python list post-processing NumPy conversion. RHS compilation (e.g. Numba) can substantially reduce overhead, although solver/event machinery remains.

Periodic Orbits and Winding Numbers

Periodicity and first return

Let Σ^+ be a transverse section with one chosen crossing direction.

- The **First Return Time** $\tau(\boldsymbol{q})$ is the first positive time that the trajectory from $\boldsymbol{q} \in \Sigma^+$ returns to Σ^+.
- The **First Return Map** records that next crossing:

$$\boldsymbol{p}(\boldsymbol{q}) = \boldsymbol{y}\big(\tau(\boldsymbol{q}); \boldsymbol{q}\big), \qquad \boldsymbol{q}_{j+1} = \boldsymbol{p}(\boldsymbol{q}_j).$$

- A periodic orbit closes in the flow. Its crossings form a periodic orbit of the map $\boldsymbol{p}$.

Once-around periodic orbit

Choose Σ^+ so the Rössler spiral crosses it once per turn.

- A once-around periodic orbit returns to the same section point after one turn:

$$\boldsymbol{p}(\boldsymbol{q}) = \boldsymbol{q}.$$

- Define the fixed-point residual

$$\boldsymbol{r}(\boldsymbol{q}) = \boldsymbol{p}(\boldsymbol{q}) - \boldsymbol{q}.$$

  The periodic orbit problem is $\boldsymbol{r}(\boldsymbol{q}) = \boldsymbol{0}$.

- This is a nonlinear equation on the section. We solve it with Newton's method, starting from a nearly returning trajectory.

Newton step from the sensitivity equation

Integrate the trajectory and its sensitivity matrix together:

$$\boldsymbol{y}' = \boldsymbol{f}(\boldsymbol{y}), \qquad \boldsymbol{S}' = D\boldsymbol{f}(\boldsymbol{y})\boldsymbol{S}, \qquad \boldsymbol{S}(0) = I.$$

- At the first return, $\boldsymbol{S}(\tau)$ gives sensitivity to the starting point. Correct it for the changing return time to obtain $D\boldsymbol{p}(\boldsymbol{q})$ on Σ^+.
- Since $D\boldsymbol{r}(\boldsymbol{q}) = D\boldsymbol{p}(\boldsymbol{q}) - I$, one Newton step is

$$\big(D\boldsymbol{p}(\boldsymbol{q}^{(\ell)}) - I\big)\Delta\boldsymbol{q} = -\boldsymbol{r}(\boldsymbol{q}^{(\ell)}), \qquad \boldsymbol{q}^{(\ell+1)} = \boldsymbol{q}^{(\ell)} + \Delta\boldsymbol{q}.$$

- Thus the same sensitivity equation used earlier supplies the derivative needed by Newton's method.

Twice around with an intermediate state

Search for both section crossings $\boldsymbol{q}_0$ and $\boldsymbol{q}_1$. Stack them into one unknown:

$$\boldsymbol{z} = \begin{pmatrix} \boldsymbol{q}_0 \\ \boldsymbol{q}_1 \end{pmatrix}$$

One return must carry each point to the other:

$$\boldsymbol{p}(\boldsymbol{q}_0) = \boldsymbol{q}_1, \qquad \boldsymbol{p}(\boldsymbol{q}_1) = \boldsymbol{q}_0$$

Define the coupled residual

$$\boldsymbol{r}_2(\boldsymbol{z}) = \begin{pmatrix} \boldsymbol{p}(\boldsymbol{q}_0) - \boldsymbol{q}_1 \\ \boldsymbol{p}(\boldsymbol{q}_1) - \boldsymbol{q}_0 \end{pmatrix} = \boldsymbol{0}$$

- This system searches for the intermediate state instead of hiding it inside $\boldsymbol{p}^2$.
- Require $\boldsymbol{q}_0 \neq \boldsymbol{q}_1$ to exclude a once-around fixed point.

Block Newton system for two returns

At Newton iteration ℓ, define the stacked correction

$$\Delta \boldsymbol{z}^{(\ell)} = \begin{pmatrix} \Delta \boldsymbol{q}_0^{(\ell)} \\ \Delta \boldsymbol{q}_1^{(\ell)} \end{pmatrix}$$

Compute $\boldsymbol{A}_j = D\boldsymbol{p}(\boldsymbol{q}_j^{(\ell)})$ from the sensitivity equation on each return. Then solve

$$\boldsymbol{J}_2 \Delta \boldsymbol{z}^{(\ell)} = -\boldsymbol{r}_2(\boldsymbol{z}^{(\ell)}), \qquad \boldsymbol{J}_2 = \begin{pmatrix} \boldsymbol{A}_0 & -I \\ -I & \boldsymbol{A}_1 \end{pmatrix}$$

- Update both crossings together: $\boldsymbol{z}^{(\ell+1)} = \boldsymbol{z}^{(\ell)} + \Delta \boldsymbol{z}^{(\ell)}$.
- Forming $\boldsymbol{A}_1 \boldsymbol{A}_0$ compounds expansion and contraction over both returns. Its condition number can be much larger than those of the one-return sensitivities.
- The block system keeps the two legs separate and is usually better conditioned.

Three times around: target $r_3(z_3) = 0$

To exclude a once-around fixed point, require $q_0 \neq q_1$. Three is prime, so the remaining cycle has minimal map period 3.

Stacked state and residual

$$z_3 = \begin{pmatrix} q_0 \\ q_1 \\ q_2 \end{pmatrix}$$

$$r_3(z_3) = \begin{pmatrix} p(q_0) - q_1 \\ p(q_1) - q_2 \\ p(q_2) - q_0 \end{pmatrix}$$

Block Newton system

$$A_j = Dp(q_j)$$

$$J_3 = \begin{pmatrix} A_0 & -I & 0 \\ 0 & A_1 & -I \\ -I & 0 & A_2 \end{pmatrix}$$

$$J_3 \Delta z_3 = -r_3(z_3)$$

Four times around: target $r_4(z_4) = 0$

To exclude once-around and twice-around cycles, require $\boldsymbol{q}_0 \neq \boldsymbol{q}_1$ and $\boldsymbol{q}_0 \neq \boldsymbol{q}_2$, respectively. The remaining cycle has minimal map period 4.

Stacked state and residual

$$\boldsymbol{z}_4 = \begin{pmatrix} \boldsymbol{q}_0 \\ \boldsymbol{q}_1 \\ \boldsymbol{q}_2 \\ \boldsymbol{q}_3 \end{pmatrix}$$

$$\boldsymbol{r}_4(\boldsymbol{z}_4) = \begin{pmatrix} \boldsymbol{p}(\boldsymbol{q}_0) - \boldsymbol{q}_1 \\ \boldsymbol{p}(\boldsymbol{q}_1) - \boldsymbol{q}_2 \\ \boldsymbol{p}(\boldsymbol{q}_2) - \boldsymbol{q}_3 \\ \boldsymbol{p}(\boldsymbol{q}_3) - \boldsymbol{q}_0 \end{pmatrix}$$

Block Newton system

$$\boldsymbol{A}_j = D\boldsymbol{p}(\boldsymbol{q}_j)$$

$$\boldsymbol{J}_4 = \begin{pmatrix} \boldsymbol{A}_0 & -I & 0 & 0 \\ 0 & \boldsymbol{A}_1 & -I & 0 \\ 0 & 0 & \boldsymbol{A}_2 & -I \\ -I & 0 & 0 & \boldsymbol{A}_3 \end{pmatrix}$$

$$\boldsymbol{J}_4 \Delta \boldsymbol{z}_4 = -\boldsymbol{r}_4(\boldsymbol{z}_4)$$

m times around: target $r_m(z_m) = 0$

To obtain minimal map period m, require $\boldsymbol{q}_d \neq \boldsymbol{q}_0$ for every proper divisor d of m.

Stacked state and residual

$$\boldsymbol{z}_m = \begin{pmatrix} \boldsymbol{q}_0 \\ \boldsymbol{q}_1 \\ \vdots \\ \boldsymbol{q}_{m-1} \end{pmatrix}$$

$$\boldsymbol{r}_m(\boldsymbol{z}_m) = \begin{pmatrix} \boldsymbol{p}(\boldsymbol{q}_0) - \boldsymbol{q}_1 \\ \boldsymbol{p}(\boldsymbol{q}_1) - \boldsymbol{q}_2 \\ \vdots \\ \boldsymbol{p}(\boldsymbol{q}_{m-1}) - \boldsymbol{q}_0 \end{pmatrix}$$

Block Newton system

$$\boldsymbol{A}_j = D\boldsymbol{p}(\boldsymbol{q}_j)$$

$$\boldsymbol{J}_m = \begin{pmatrix} \boldsymbol{A}_0 & -I & 0 & \cdots & 0 \\ 0 & \boldsymbol{A}_1 & -I & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \boldsymbol{A}_{m-2} & -I \\ -I & 0 & \cdots & 0 & \boldsymbol{A}_{m-1} \end{pmatrix}$$

$$\boldsymbol{J}_m \Delta \boldsymbol{z}_m = -\boldsymbol{r}_m(\boldsymbol{z}_m)$$

Multiple returns

The mth return is the composition

$$\boldsymbol{p}^m = \underbrace{\boldsymbol{p} \circ \boldsymbol{p} \circ \cdots \circ \boldsymbol{p}}_{m \text{ times}}.$$

- A point has **Minimal Map Period** m when

$$\boldsymbol{p}^m(\boldsymbol{q}) = \boldsymbol{q}, \qquad \boldsymbol{p}^j(\boldsymbol{q}) \neq \boldsymbol{q} \quad (1 \leq j < m).$$

- The corresponding flow orbit closes after m selected crossings. Its elapsed period is

$$T = \sum_{j=0}^{m-1} \tau\big(\boldsymbol{p}^j(\boldsymbol{q})\big).$$

- m counts returns. T measures time.

Turns around the Rössler spiral

Choose a vertical half-plane Σ^+ through the origin and select one crossing direction.

- A trajectory near the familiar Rössler spiral crosses this half-plane once on each trip around the spiral.
- Here "winding number" means the number of these turns before the periodic orbit closes. We use the term only for this picture.
- A fixed point of $\boldsymbol{p}$ goes around once. A point of minimal map period m goes around m times.
- Minimal map periods $m = 2, 4, 8$ therefore give two-, four-, and eight-turn Rössler orbits.

Periodic orbits as nonlinear equations

For a parameter c, define the periodic-point residual

$$\boldsymbol{r}_m(\boldsymbol{q}, c) = \boldsymbol{p}_c^m(\boldsymbol{q}) - \boldsymbol{q}.$$

- Solve $\boldsymbol{r}_m(\boldsymbol{q}, c) = \boldsymbol{0}$ on Σ^+.
- Reject roots of $\boldsymbol{r}_j = \boldsymbol{0}$ for divisors $j < m$. This enforces minimal map period m.
- Recover every crossing from $\boldsymbol{q}_j = \boldsymbol{p}_c^j(\boldsymbol{q})$, $j = 0, \ldots, m-1$.
- Integrate between consecutive crossings to reconstruct the periodic orbit in the full Rössler state space.

Computing the 2-, 4-, and 8-turn orbits

1: Choose $m \in \{2, 4, 8\}$ and a parameter value c
2: Use section data to choose an initial guess $\boldsymbol{q}^{(0)}$
3: **for** $\ell = 0, 1, 2, \ldots$ **do**
4: Evaluate $\boldsymbol{p}_c^m(\boldsymbol{q}^{(\ell)})$ by event-driven integration
5: Form $\boldsymbol{r}_m = \boldsymbol{p}_c^m(\boldsymbol{q}^{(\ell)}) - \boldsymbol{q}^{(\ell)}$
6: Solve $D\boldsymbol{r}_m \, \Delta\boldsymbol{q} = -\boldsymbol{r}_m$
7: $\boldsymbol{q}^{(\ell+1)} \leftarrow \boldsymbol{q}^{(\ell)} + \Delta\boldsymbol{q}$
8: **end for**

Continue the solution with minimal map period 2 in c, then use each doubling to initialize the solves for minimal map periods 4 and 8.

Stability on the section

At a periodic point $\boldsymbol{q}$ with minimal map period m, linearize the mth return:

$$D(\boldsymbol{p}_c^m)(\boldsymbol{q}) = D\boldsymbol{p}_c(\boldsymbol{q}_{m-1}) \cdots D\boldsymbol{p}_c(\boldsymbol{q}_1) D\boldsymbol{p}_c(\boldsymbol{q}_0).$$

- Its eigenvalues are the multipliers of the periodic point.
- The orbit is attracting on the section when every multiplier has magnitude less than one.
- The section removes the neutral flow direction. We test the dynamics transverse to the periodic orbit.

Period doubling

Suppose an orbit of the return map with minimal map period m varies with the parameter c.

- A **Period-Doubling Bifurcation** occurs when a multiplier of $D(\boldsymbol{p}_c^m)$ passes through -1.
- The original orbit with minimal map period m changes stability. A nearby orbit of the return map with minimal map period $2m$ appears under the usual nondegeneracy conditions.
- Repeated doublings produce the sequence

$$m = 1 \longrightarrow 2 \longrightarrow 4 \longrightarrow 8 \longrightarrow \cdots .$$

- Numerically, continue each branch in c and monitor its multipliers. Use the doubled crossing pattern as the next guess.

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